METRIC

MIL-C-63982A(AR)
26 August 1992
SUPERSEDING
DOD-C-63982
29 October 1982

MILITARY SPECIFICATION

CARTRIDGE, 30MM HIGH EXPLOSIVE DUAL PURPOSE, M789

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains the requirements, quality assurance provisions, and packaging for one type of electric primed cartridge designated Cartridge, 30MM HEDP, M789 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification for
DOD-C-63980	-Cartridge, 30MM, Target Practice M788

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109	-Quality Assurance Terms, and Definitions
MIL-STD-414	-Samples Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-810	-Environmental Test Methods
MIL-STD-1167	-Ammunition Data Cards
MIL-STD-1168	-Lot Numbering of Ammunition

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.15)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

9390627	-Barrel, Test, 30MM (Aden Barrel)
9390748	-Barrel, Test: Gun M230 (Progressive w/6.5° Exit Twist)
11825879	-Link, Cartridge 30MM - M29
11825890	-Container, Shipping and Storage 30MM, Bulk Packed
11825891	-Container, Shipping and Storage 30MM, Linked, RH-Packed
11825892	-Container, Shipping & Storage, 30MM Linked, LH Packed
11825893	-Marking, Shipping and Storage Container
11825947	-Cartridge 30MM TP M788
11825954	-Cartridge, 30MM, HEDP, M789
12902718	-Packing and Marking of Ammunition Container, 30MM, M592 for 30MM Cartridges
12902719	-Strip, Ammunition, Metallic

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PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 -Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

Code of Federal Regulations

Title 49 -Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

(Copies of other Government documents, drawings and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General.

3.1.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.1.2 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg. 11825954), all associated drawings, and with all requirements specified in applicable specifications.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the Contracting Officer and no deviation from that process shall be made without prior approval (see 6.3).

3.3 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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3.4 Projectile extraction. The projectile, when subjected to a bullet pull using a chamber simulating ring at a rate of 25mm per minute ±10 percent, shall have a retention force such that the mean value less three times the standard deviation shall be greater than 8 kilo-Newtons.

3.5 Fuze torque. The projectile shall withstand a torque as specified by Dwg. 11825955.

3.6 Electrical resistance, primed case. The electrical resistance of the primed case shall be as specified on Dwg. 11825901.

3.7 Propellant charge. The propellant charge used in each cartridge shall be maintained to meet the ballistic requirements.

3.8 Muzzle velocity. The projectile shall have a mean muzzle velocity of 795-815 meters per second (m/s) at 21 degrees C ± 5 degrees C. The standard deviation shall not exceed 25 (m/s) at temperatures of 21 degrees C ± 5 degrees C and 71 degrees C. Following the exposure to the environmental conditions of 3.10, the mean velocity at 21 degrees C ± 5 degrees C shall not increase nor decrease from the mean muzzle velocity before exposure at 21 degrees C ± 5 degrees C by more than 30 m/s.

3.9 Case mouth pressure. The mean peak pressure shall not exceed 295 MPa at 21°C ± 5°C. The mean peak pressure shall not exceed 415 MPa at temperatures of -54°C and 71°C, respectively. Following environmental exposure, the mean peak pressure shall not increase or decrease from the mean peak pressure at 21°C ± 5°C by more than 60 MPa.

3.10 Environmental conditions. The cartridge shall not be damaged, deteriorated, or exhibit degraded performance beyond the limits of this specification after exposure to the following environmental conditions.

3.10.1 Temperature, humidity and altitude. Temperature/humidity/altitude cycling conditions in accordance with MIL-STD-810, Method 518, with the temperature ranging from -54°C to 65°C, relative humidity up to 95 percent, and pressure conditions equivalent to 15.2 kilometers altitude.

3.10.2 Salt fog. Exposure to salt sea atmosphere specified in MIL-STD-810, Method 509.

3.10.3 Aircraft vibration. Vibration as specified in MIL-STD-810, Method 514 except as modified in 4.5.7.3.

3.10.4 Waterproofness. Immersion for one hour in water under a pressure of 0.1 MPa ± 10% at 21°C.

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3.11 Accuracy. The cartridge when fired in a single shot test barrel (Dwg 9390627 shall be such that the standard deviation of the impact points about the mean impact point shall not exceed 1.35 mil.

3.12 Action time. The mean action time plus 3 standard deviations shall not exceed 4.0 milliseconds (ms) at -54°C 21°C and 71°C. The maximum individual action time shall not exceed 4.0 ms at -54°C ± 5°C, 21°C ± 5°C and 71°C ± 5°C. The action time is defined as the time period between the application of electric current to the primer and the exiting of the projectile from the muzzle.

3.13 Function and casualty. The cartridge shall function as required and there shall be no separation or breakup of cartridge parts during the firing sequence. In addition when fired in the automatic gun there shall be no gun stoppages caused by the ammunition. See 6.11.

3.14 Penetration.

3.14.1 Dynamic penetration. When fired from Barrel, Test: Gun M230 (Progressive w/6.5° Exit Twist) (Dwg. 9090748) the cartridge shall completely penetrate the target specified in 4.5.14 with the reliability of 70% (complete penetration is evidenced by an exit hole at the rear of the target).

3.14.2 Static penetration. Projectiles shall be tested against 3" (76.2mm) thick rolled homogeneous armor at spin rate of 660 ± 20, 775 ± 20 and 908 ± 20 revolutions per second. The respective mean penetration depths shall exceed 1.70, 1.47 and 1.37 inches."

3.15 Normal impact function. When fired against 1.6MM (nominal thickness) 2024T3 aluminum alloy sheet the cartridge shall function high order.

3.16 Contamination. Particular care should be exercised to avoid contamination of primers and propellant by oil or grease. Such contamination may adversely affect performance. Particular care should also be exercised to avoid contamination of primers and propellant by dust and other foreign matter. If contamination by foreign matter or dust is evident, the quantity shall not exceed the requirements set forth in Drawing 11825946.

3.17 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specification and the following:

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3.17.1 Process defects. The cartridges shall be free of defects such as folds, wrinkles, buckles, splits, severe scratches, cuts, dents, perforations and other forms of substandard workmanship.

3.17.2 Cleanliness. The cartridge and its components shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (See 4.3).

b. Quality conformance inspection (See 4.4).

4.3 First article inspection.

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed in Table I in the quantities indicated.

4.3.2 Inspection to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and test specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 4	DRAWING NUMBER
	Cartridge and Components		See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Liner - Proj., 30MM, HEDP, M789 (Dwg. 11825960)			
	Examination for defects		3.1.2	4.4.2.2
	Retainer, Proj., 30MM, HEDP, M789 (Dwg. 11825961)			
	Examination for defects		3.1.2	4.4.2.3
	Pellet, Explosive, Proj., 30MM, HEDP, M789 (Dwg. 11825959)			
	Examination for defects		3.1.2	4.4.2.4
	Pellet, Booster, Proj., 30MM, HEDP, M789 (Dwg. 11825962)			
	Examination for defects		3.1.2	4.4.2.5
	Charged Body Assembly Proj., 30MM, HEDP, M789 (Dwg. 11825956)			
	Examination for defects		3.1.2	4.4.2.6
	Static Penetration		3.14	4.5.14
	Flashtube Assembly (Dwg. 11825903)			
	Examination for defects		3.1.2	4.4.2.7

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE		SHEET 2 OF 4	DRAWING NUMBER
	Cartridge and Components			See Below
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Case Cartridge Assembly (Dwg. 11825091)			
	Examination for defects		3.1.2	4.4.2.8
	Projectile, 30MM, HEDP, M789 (Dwg. 11825955)			
	Examination for defects		3.1.2	4.4.2.9
	Cartridge, 30MM, HEDP, M789 (Dwg. 11825954)			
	Examination for defects		3.1.2	4.4.2.1

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge and Components	3 OF 4	11825954
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
See 1/				
A	Projectile Extraction	20, 0-1	3.4	4.5.8
A	Propellant Charge	20, 0-1	3.7	4.5.9
A	Propellant Contamination	20, 0-1	3.16	4.5.17
B	Velocity 2/	60	3.8	4.5.2
B	Case Mouth Pressure 2/	60	3.9	4.5.3
B	Action Time 2/	60	3.12	4.5.4
B	Accuracy	60	3.11	4.5.10
B	Function & Casualty (Mann Barrel)	60, 0-1	3.13	4.5.5
	Temperature/Humidity/Altitude	20, 0-1	3.10.1	4.5.7.1
	Salt Spray	20, 0-1	3.10.2	4.5.7.2
	Aircraft Vibration	20, 0-1	3.10.3	4.5.7.3

NOTES: 1/ Tests identified by the same letter shall be conducted on the same samples.
2/ 40 cartridges at 21°C and 10 cartridges each at -54°C & 71°C. Data collected at -54°C and 71°C will be recorded for information only.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge and Components	4 OF 4	11825954
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Dynamic Penetration 3/	25, 4-5	3.14	4.5.14
	Normal Impact Function 3/	45, 1-2	3.15	5/
	Waterproofness	20, 0-1	3.10.4	4.5.7.4
	Function & Casualty (Auto gun) 4/	200, 0-1	3.13	4.5.5

NOTES: 3/ Fired at 21°C.
4/ Gun specified in the contract or purchase order.
5/ The target shall be placed 200 meters from the muzzle. Use the M230 Mann Barrel.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Component parts. Unless otherwise specified, component parts shall be homogeneous and of a quantity convenient to the contractor, except that each cartridge lot shall contain:

a. Cartridge cases from one unchanged process, from one lot interfix number and one manufacturer.

b. Primers from one lot and one manufacturer.

c. Projectile components from one unchanged process, from one lot interfix number, and one manufacturer.

d. Propellant from one lot and one manufacturer.

4.4.1.2 Assembled cartridge lot. Each lot of ammunition shall be identified in accordance with MIL-STD-1168, supplemented as directed by the procuring activity. Each packed lot of ammunition shall further be identified by a national stock number assigned by the procuring activity.

4.5.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150000 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.17). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.1	Cartridge, 30MM, HEDP M789 1/		11825954
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical: None defined				
Major				
101	Case to projectile crimp missing	Level III	3.1.2	Visual
102	Split or perforated case	Level III	3.1.2	Visual
103	Exterior profile (max) 2/	Level III	3.1.2	Gage
104	Overall length (max)	Level III	3.1.2	Gage
105	Case crimp diameter (max) 3/	Level III	3.1.2	Gage
106	Max gap case-projectile interface 5/	Level III	3.1.2	Gage
107	Case projectile diameter between crimps	Level III	3.1.2	Gage/Visual
108	Fuze nose cap damaged	Level III	3.1.2	Visual

NOTES:
* Test may be conducted using the same sample cartridges.
** Test may be conducted using the same sample cartridges.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, 30MM, HEDP M789 1/	2 OF 3	11825954
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Tests			
109	Projectile extraction *	Table III 0-1	3.4	4.5.8
110	Presence of waterproof seal * (projectile to case)	Table III 0-1	3.1.2	Visual
111	Propellant contamination *	Table III 0-1	3.16	4.5.9
112	Propellant charge & propellant weight * 4/	Table III 0-1	3.7	4.5.9
	Ballistic performance (21°C)			
113	Velocity **	Table IV	3.8	4.5.2
114	Case Mouth Pressure **		3.9	4.5.3
115	Action time **		3.12	4.5.4
116	Accuracy **	25	3.11	4.5.10
117	Normal impact function 7/	90, 2-11	3.15	4.5.15
118	Function & Casualty	Table VII 0-1		
	Ballistic performance (-54°C & 71°C)			
119	Velocity **	Table V	3.8	4.5.2
120	Case Mouth Pressure **		3.9	4.5.3
121	Action time **		3.12	4.5.4

NOTES:

** Test may be conducted using the same sample cartridges.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, 30MM, HEDP M789 1/	3 OF 3	11825954
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
122	Function & Casualty (Mann Barrel)	Table VII	3.13	4.55
Minor				
201	Protective coating damaged or incomplete 6/	Level IV	3.1.2	Visual/6.12
202	Incorrect type cartridge	Level IV	3.1.2	Visual
203	Foreign material (except corrosion)	Level IV	3.1.2	Visual
204	Workmanship	Level IV	3.17	Visual
205	Folds, wrinkles, buckles, deep scratches, cuts and dents	Level IV	3.1.2	Visual
206	Corrosion	Level IV	3.1.2	Visual
207	Marking incorret, illegible or missing	Level 1V	3.1.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-63982A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Liner, Proj., 30MM, HEDP M789	1 OF 1	11825960
			NEXT HIGHER ASSEMBLY 11825956

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical: None defined				
Major				
101	Diameter, inside at surface "X"	Level III	3.1.2	Gage
102	Diameter, outside at surface "X"	Level III	3.1.2	Gage
103	Large outside diameter, max.	Level III	3.1.2	Gage
104	Large inside diameter, min.	Level III	3.1.2	Gage
105	Concentricity of outside diameter small ends	Level III	3.1.2	Gage
106	Location of large datum diameter	Level III	3.1.2	Gage
107	Location of small datum diameter	Level III	3.1.2	Gage
108	Thickness T, section A-A & B-B	Level III	3.1.2	Gage
Minor:				
201	Overall length	Level V	3.1.1	Gage
202	Large outside diameter, min	Level V	3.1.1	Gage
203	Large inside diameter, max	Level V	3.1.1	Gage
204	Workmanship	Level V	3.17	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 11825961
4.4.2.3	Retainer, Proj., 30MM, HEDP M789		NEXT HIGHER ASSEMBLY 11825956

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major				
101	Outside diameter	Level V	3.1.2	Gage
102	Height of retainer	Level V	3.1.2	Gage
103	Thread diameter	Level V	3.1.2	Gage
Minor:				
201	Length of thread	Level VI	3.1.2	Gage
202	Evidence of poor workmanship	Level VI	3.17	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-63982A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Pellet, Explosive, Proj., 30MM, HEDP M789		11825959
			NEXT HIGHER ASSEMBLY 11825956

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major				
101	Weight	Level III	3.1.2	Scale/Balance
Minor:				
201	Outside diameter	Level V	3.1.2	Gage
202	Length	Level V	3.1.2	SMTE 1/

NOTES: 1/ Standard measurement test equipment.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-63982A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Pellet, Booster, Proj., 30MM, HEDP M789	1 OF 1	11825962
			NEXT HIGHER ASSEMBLY 11825956

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major				
101	Weight	Level III	3.1.2	Scale/Balance
Minor:				
201	Outside diameter	Level V	3.1.2	Gage
202	Length	Level V	3.1.2	SMTE

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-63982A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Charged Body Assembly Proj., 30MM, HEDP, M789	1 OF 1	11825956
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major				
101	Min depth and presence of retainer	Level II	3.1.2	Gage
102	Presence of seal	Level II	3.1.2	Visual
103	Explosive of threads at indicated length	Level II	3.1.2	Visual
104	Weight	Level II	3.1.1	Gage
Minor:				
201	Workmanship	Level IV	3.16	

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Flashtube Assembly	1 OF 1	11825903
			NEXT HIGHER ASSEMBLY 11825901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major				
101	Ignitor material missing or incorrect (Broken pellets permissible)	Level II	3.1.2	Visual/Weight 1/
Minor:				
201	Evidence of poor workmanship	Level IV	3.17	Visual

NOTES: 1/ Weight Method: Determine minimum flash tube/tape weight. Weigh each flash tube assembly and accept assemblies with a weight equal to or greater than the maximum pellet weight allowed plus the minimum flash tube/tape weight.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER 11825901
4.4.2.8	Case Cartridge Assembly			NEXT HIGHER ASSEMBLY 11825954
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major				
101	Depth of primer seating	Level II	3.1.2	Gage
102	Primer/Case Resistance	Level II	3.1.2	Gage
103	Presence of Igniter Tape Seal	Level II	3.1.2	Gage
104	Dimpling of Cartridge Case (see note)	Level II	3.1.2	Visual
105	Loose Primer	Level II	3.1.2	Manual
106	Corrosion	Level II	3.1.2	Visual
Minor:				
201	Primer Waterproofing Missing or incomplete	Level III	3.1.2	Visual
202	Incomplete or Missing Primer Crimp	Level III	3.1.2	Visual
203	Evidence of Poor Workmanship	Level IV	3.16/ 3.17	Visual

NOTES:

Note: Inspection shall be performed at cartridge level.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-63982A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Projectile, 30MM, HEDP, M789	1 OF 1	11825954
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major				
101	Disassembly torque test of fuze	Level II	3.1.2/ 3.5	4.5.16
102	Gap between fuze and body, max.	Level II	5/	Gage
103	Bourrellet diameter, max.	Level II	3.1.2	Gage
Minor:				
201	Workmanship	Level IV	3.17	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-63982A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Container, Shipping and Storage, 30MM Packed	1 OF 1	11825891 or 11825890 or 11825892
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major		1/		
101	Cartridge inverted	100%	3.1.2	Visual
102	Cushions, fillers, liners and separators missing	100%	3.1.2	Visual
103	Tubes missing (if applicable)	100%	3.1.2	Visual
104	Link orientation (LH-RH) (if applicable)	100%	3.1.2	Visual
105	Contents loose	100%	3.1.2	Visual
106	Link improper (anyone of the four (4) tabs of link not engaged, link inverted; link damaged; link distorted)	100%	3.1.2	Visual
Minor:				
201	Metallic seal missing	100%	3.1.2	Visual

NOTES:

1/ Sample size for all Majors and Minors, 5 boxes/lot.

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-63982A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Marking on Shipping and Storage Container M592, 30MM Cartridge	1 OF 1	11825893
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major:	None defined			
Minor:				
201	Marking misleading or unidentifiable	100%	3.1	Visual

NOTES:

AMSMC Form 1570a, 1 Jul 89

Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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NOTES FOR PARAGRAPH 4.4.2.1 & 4.4.2.8

1/ Visual standards/machine masters established and approved by the Government as defined in 6.4, shall be used in evaluating cartridge case and cartridge visual defects. Major components shall have previously been accepted in accordance with the acceptance criteria defined in the specification listed in the applicable drawing.

2/ A chamber gage in accordance with Dwg. 9354298 will be used. The maximum force to insert the cartridge into the gage shall comply with either of the following criteria.

a. Apply a static force of not to exceed 270 Newtons.

b. Apply a dynamic force by hand (no mechanical aid) which does not damage the cartridge.

3/ The lot shall not be realized during inspection verification for oversize conditions provided that:

a. The crimp is visually complete for 360° and,

b. A sample quantity of the discrepant cartridges in accordance with Table II meet the projectile extraction requirement of 3.4.

4/ A cartridge weighing less than the value determined by the formula specified in 3.16 shall be rejected. The 100% inspection method may be by verification of weight of each cartridge or by probe for a fixed minimum propellant height in a charged cartridge case. If the probe method is used, the following method shall be used to establish the minimum height for each cartridge lot:

a. Establish the charge weight in accordance with 4.5.9.

b. Prior to any loading charge a minimum of 25 cases to the specified weight, measure each case and calculate the mean propellant height for the propellant batch to be loaded.

c. Repeat step b with a propellant charge weight of 1 gram less than the established mean charge weight.

d. Adjust the probe height such that any cartridge in which the propellant charge is 1 gram below the mean weight will be rejected.

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e. Verify the probe height at the beginning of production and every 4 hours of production or portion thereof, and at the end of production of the lot. Any probe light failure shall cause rejection of the cartridges loaded since the last acceptable probe height verification.

f. Verify the proper propellant height of the end of lot production by repeating steps b and c and comparing the established probe height.

5/ A gap of 0.25mm or less at any point on the circumference is acceptable.

6/ The lot shall not be penalized for anodize crazing in the cartridge crimp grooves.

7/ The failure of two or less projectiles to function high order detonation shall accept the lot. If 3 - 10 failures occur, the lot will be retested per 4.5.15. Failure of 11 or more projectiles shall be cause for rejection of the lot.

TABLE III. Sample size for projectile extraction test.

Lot Size	Sample Size
0 - 800	10
801 - 1,300	15
1,301 - 3,200	20
3,201 - 8,000	25
8,001 - 22,000	30
22,001 and over	35

Reference: MIL-STD-414, Level 2, Table A-2
Except for lots of less than 800 units.

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TABLE IV. Sample size for 21°C ballistic tests.

Lot Size	First Sample Size	Second Sample Size
0 - 800	30	35
801 - 1,300	35	40
1,301 - 3,200	40	50
3,201 - 8,000	50	75
8,001 - 22,000	75	100
22,001 - 110,000	100	140
110,000 - 550,000	150	190
Greater than 550,000	200	255

Reference: MIL-STD-414, Level 2, Table A-2
Except for lots of less than 800 units.

TABLE V. Sample size for 71°C and -54°C ballistic tests.

	Sample Size for each Temperature	
Delivered Lot Size	First Sample Size	Second Sample Size
0 - 800	10	10
801 - 1,300	10	10
1,301 - 3,200	10	15
3,201 - 8,000	15	20
8,001 - 22,000	20	20
Greater than 22,000	25	25

Reference: MIL-STD-414, Level 2, Table A-2
Except for lots of less than 3200 units.

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TABLE VI. Firing defects 1/

Item	Defect	Acc	Rej
Cartridge	Misfire due to cartridge 2/	0	1
	Failure to chamber due to Cartridge 3/	0	1
	Failure to extract due to cartridge 4/	0	1
	Projectile remaining in bore 5/	0	1
	Early burst 6/	0	1
Primer	Blown primer 7/	0	1
	Loose primer 8/	0	1
	Internal minor primer leak 9/	1	2
	External minor primer leak 9/	1	2
	Primer leak 9/	0	1
Cartridge Case	Split or rupture in the cartridge case 10/	0	1
	Detached metal from the cartridge case 11/	0	1
	Erosion, internal or external 12/	1	2
	Burn-through, base 13/	0	1
	Burn-through, major 13/	0	1
	Burn-through, minor 13/	1	2
Projectile	Metal parts separation 14/	0	1

1/ These defects apply only to cartridges fired in the approved test weapon.

2/ Misfire. If a misfire occurs, a second firing voltage application is not allowed unless a non-ammunition cause for the misfire is confirmed and corrected.

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3/ Failure to chamber. The cartridge cannot be inserted into chamber sufficient distance to allow the breach to be engaged or cartridge cannot be bottomed by the breach within the limits of an axial load of 270 Newtons.

4/ Failure to extract. After firing, the cartridge case cannot be removed without failure of the extractor ring. Foreign debris at cartridge/chamber interface causing cartridge case removal difficulty does not constitute a valid failure.

5/ Projectile remaining in bore. This defect is defined as a failure of the projectile to exit the barrel after the primer has fired.

6/ Early burst. An early burst is considered to have occurred if the projectile functions prior to impact.

7/ Blown primer. The cartridge case shall be visually examined. A blown primer is a primer which when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and the primer pocket are enlarged and deformed.

8/ Loose primer. The cartridge case shall be visually examined. A loose primer is defined as a looseness which permits the fired primer to fall from the primer pocket after the cartridge has been fired, without distortion of the primer pocket.

9/ Primer leaks. Primer leads shall be defined as follows:

a. Suspect. A suspect primer leak is defined as the definite evidence of smudging or discoloration around any part of the periphery of the primer cup, button, insulator, or primer pocket. The fired primer must be removed by sectioning the cartridge case and must be disassembled to determine whether actual gas leakage has occurred.

b. Internal minor leak. A suspect primer which, upon disassembly shows definite evidence of a leakage path between primer components or through a crack or split in any primer component.

c. External minor primer leak. A suspect primer, which upon disassembly, shows definite evidence of a leakage path between the primer cup and primer pocket without evidence of a leakage path between the primer cup and primer pocket without evidence of erosion of the pocket.

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d. Major leak. A Primer leak which shows erosion of any primer component.

e. Critical leak. A primer leak which shows evidence of erosion of the primer pocket or melting of any primer component.

f. A dislodged primer button is not cause for rejection.

10/ Split or ruptured cartridge case. The cartridge case shall be visually examined. A split or rupture is acceptable providing proper obturation is obtained.

11/ Detached metal. The cartridge case shall be visually examined. Metal sheared or missing from the exterior profile of a fried cartridge case, such as a rim or neck shear or ignition system component, shall be classed as detached metal. Shavings or metal from the interior wall of the neck in the crimp area shall not be considered as detached metal.

12/ Mouth erosion. Cartridge case mouth erosion is defined as erosion of an area greater than 160 square millimeters to a depth such that the remaining thickness of the cartridge case mouth is reduced to less than 0.25mm.

13/ Burn-through.

Major. Major burn-through is defined as a hole through the case due to melting which exceeds 2.5mm in any dimension.

Minor. Minor burn-through is defined as a hole through the case due to melting which is 2.5mm or less in any dimension.

Base. Base burn-through is defined as a hole in the base area not to include primer leaks. The base area is defined as the area of the case not supported by the chamber.

14/ Loss of projectile structual integrity (metal parts only). See 4.5.6

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TABLE VII. Function and casualty firing defect accept/reject criteria

Lot Size	AQL			
	0.065	0.25	0.40	0.65
501 - 800	0/1 *	0/1 *	0/1 *	0/1 *
801 - 1,300	0/1 *	0/1 *	0/1 *	0/2 1/2
1,301 - 3,200	0/1 *	0/1 *	0/2 1/2	0/2 2/3
3,201 - 8,000	0/1 *	0/1 *	0/2 2/3	0/3 4/5
8,001 - 22,000	0/1 *	0/2 2/3	0/3 3/4	1/3 4/5
22,001 - 110,000	0/1 *	0/2 2/3	0/3 4/5	1/4 6/7
110,000 - 550,000	0/1 *	0/3 3/4	1/4 4/5	2/5 6/7
Over 550,000	0/1 *	1/3 3/4	2/4 6/7	3/6 8/9

* Indicates single sampling plan, i.e., no retest

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4.4.3 Testing.

4.4.3.1 Unlisted firing defects. The lot should be suspended and referred to the contracting officer for disposition if any malfunction or casualty not covered by this specification occurs in any firing test.

4.4.3.2 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.14 herein.

4.5 Methods of inspection.

4.5.1 Test barrel. Test samples for all firing test shall be fired from a test barrel conforming to Drawing 9390627, unless otherwise specified. (See 6.6 for criteria for barrel replacement.)

4.5.2 Muzzle velocity. Velocity shall be measured by the use of velocity coils or light screens. The test cartridges shall be conditioned at -54°C, 21°C and 71°C respectively before firing. The mean velocity and standard deviations from the mean of the sample cartridges in each group, shall conform to 3.8. The velocity shall be corrected to the muzzle velocity as follows:

$$V_m = \frac{d_1}{t} + c\,[d_2 + 0.5d_1]$$

Where:

t = time between screens, seconds

d_1 = distance between screens, meters

d_2 = distance from muzzle to first screen, meters

c = constant of proportionality, 0.75 meters/second/meter

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4.5.3 Case mouth pressure. The test rounds shall be conditioned at -54°C, 21°C and 71°C temperatures respectively and fired from a test fixture barrel. Pressure measurements shall be made at the case mouth with the piezo-electric transducer. A permanent record shall be made of the resulting pressure. Data at -54°C and 71°C will be recorded. This record may be a pressure trace, paper print out of the data or other similar instrument reading. The pressure shall conform to the requirements of 3.9.

4.5.4 Action time. The test rounds shall be conditioned at -54°C, 21°C and 71°C and fired from a test fixture barrel. The time between application of voltage to primer and exit of the projectile from the muzzle shall be measured. The time shall comply with requirement of 3.13. Any corrected action time of 2.40 milliseconds or less shall be disregarded as an instrumentation error.

4.5.5 Function and casualty. These tests shall be conducted, using the test methods prescribed in AMCR-715-505, Volume 8, Chapter 1, Section 3 as a guide. Witness panels for structural integrity shall be placed approximately 5 meters from the muzzle. A failure is characterized by observance of any of the firing defects listed in Table V. Test rounds, when required, shall be fired in the M230 Gun. Test results shall conform to the requirements of 3.12.

4.5.6 Projectile structural integrity. Projectile structural integrity shall be verified in conjunction with the ballistic tests of 4.5.2, 4.5.3 and 4.5.4. A witness board/card shall be located approximately 5 meters from the gun muzzle. Loss of projectile structural integrity is defined as projectile fracture, break-up, rotating band failure as defined in 6.13 or loss of parts as evidenced by microflash photography hole(s) in the witness board or recovery of fragments.

4.5.7 Environmental tests. Unless otherwise specified, performance after environmental exposure, as specified in the applicable test procedure, shall consist of conducting the tests of 4.5.2, 4.5.3 and 4.5.4, except all test rounds shall be conditioned at 21°C prior to firing.

4.5.7.1 Temperature, humidity and altitude. The cartridge shall be subjected to the temperature/humidity/altitude of MIL-STD-810, Method 509 for a period of 48 hours. After exposure, the test item shall be subjected to the tests referenced in 4.5.7.

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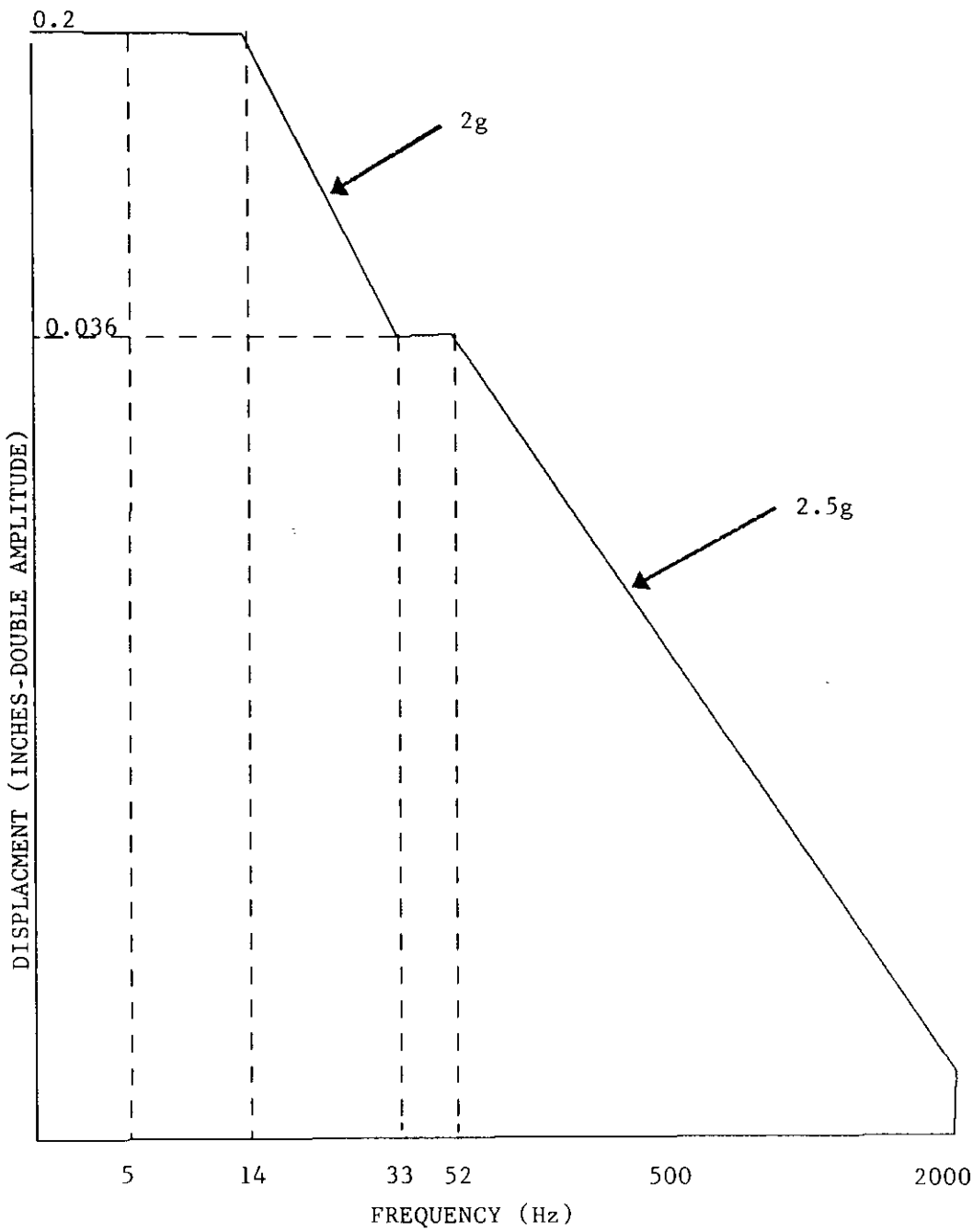


FIGURE 1. Vibration Test Curves for Equipment Installed in Helicopters, Equipment Category c. (Ref. Figure 5.14.2-3 of MIL-STD-810C)

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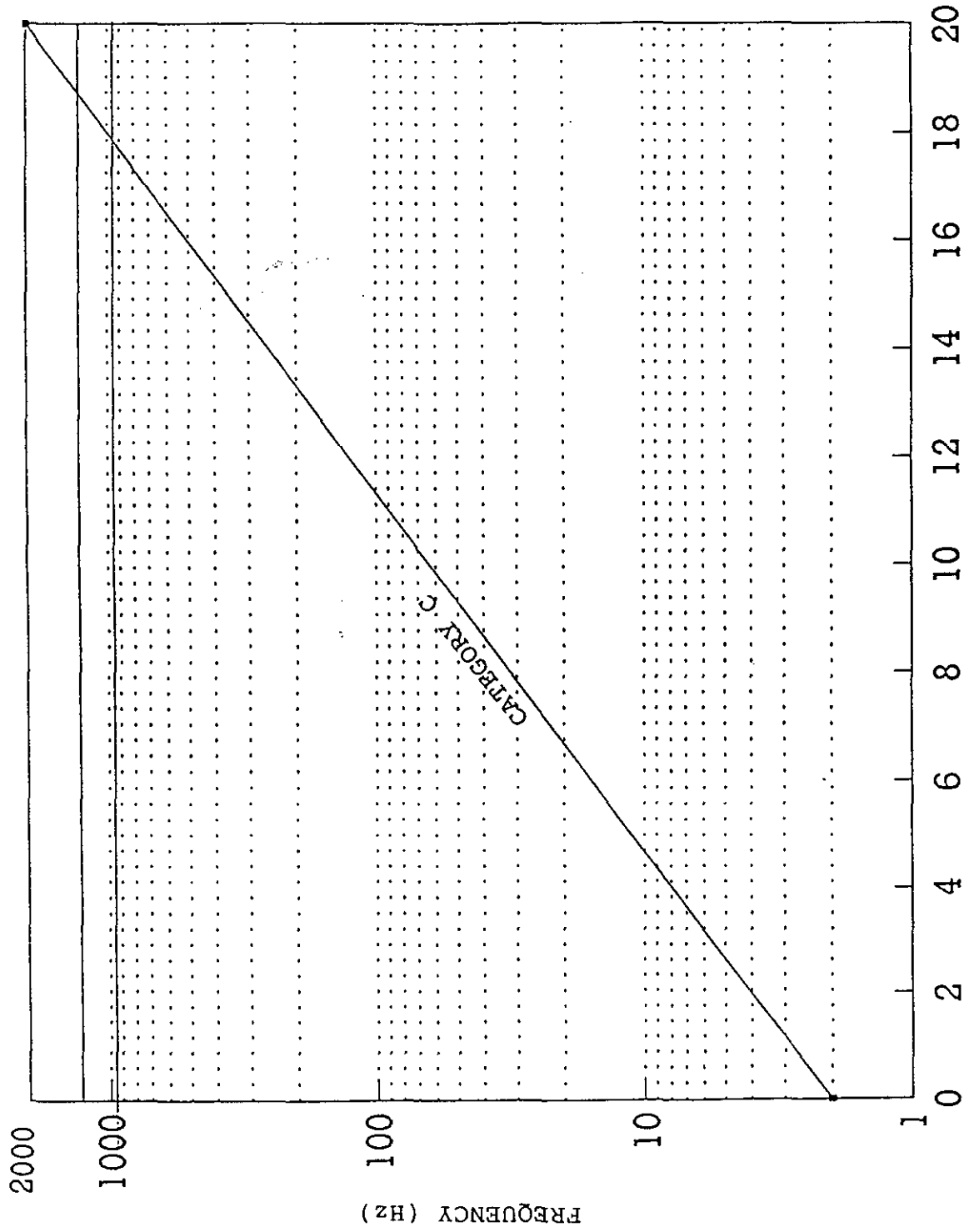


FIGURE 2. Logarithmic Sweep. (Ref. Figure 514.2-10

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4.5.7.2 Salt fog. The cartridge shall be subjected to the salt fog test of MIL-STD-810, Method 509 for a period of 48 hours. After exposure, the test item shall be subjected to the tests referenced in 4.5.7.

4.5.7.3 Aircraft vibration. The ammunition shall be subjected to a broad band random vibration profile in accordance with the requirements of MIL-STD-810, Method 514.2, Procedure I, Category C. The test will be conducted with an M592 Ammo Can. The vibration spectrum shall be as shown in Figures 1 and 2. The test item shall be mounted to a test fixture simulating a feed system in a normal airplane mounting. The test duration shall be 45 minutes in each of three mutually perpendicular axis at each of two temperature condition, $-40^{o} \pm 2.8^{o}C$ and $65.5 \pm 2.8^{o}C$. Following exposure, the test items shall be subjected to the tests referenced in 4.5.7.

4.5.7.4 Waterproofness. Sample cartridges shall be immersed for one hour in water at the temperature and gauge pressure specified in 3.10.4. The cartridges shall be taken out and wiped dry. The test quantity shall then be tested in accordance with 4.5.7 to verify conformance with the requirements of 3.10.

4.5.8 Projectile extraction. The base of the cartridge case shall be secured in a holding fixture and a ring simulating the gun chamber shall be installed in the case neck area. A clamp shall be applied at a constant strain rate of 25MM per minute $\pm$ 10% until the projectile is withdrawn from the cartridge case. The peak axial load shall be recorded and compared to the requirements of 3.4. A case neck separation during this test shall be cause for rejection. A split of greater than 5MM observed during projectile extraction shall be cause for rejection.

4.5.9 Propellant charge. The charge of propellant from a given lot, which is required to meet the performance requirement of 3.8, 3.9 and 3.12 shall be determined by ballistic test. The minimum charge weight assigned shall be used in determining the minimum cartridge weight in accordance with 6.8.

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4.5.10 Accuracy. The test shall be conducted using a single shot test barrel with cartridge conditioned to 21°C. The target shall be positioned at 200 ft ± 6.8 ft from the gun muzzle. After three barrel fouling shots have been fired the test shots shall be fired without moving the gun or target. The target impacts shall be measured for dispersion from the calculated center of impact as described in AMCR 715-505, Vol 8, Chapter I, Section 4.4. The impact dispersion of these shots will comply with the requirements of 3.11. Testing shall not be conducted if wind velocity including gusts, is 10 knots or more. Alternate procedure at 25 m ± 1 m may be used with approved automated electronic accuracy systems (see 6.14).

4.5.11 Reference rounds. Reference rounds shall be fired in conjunction with ballistic and accuracy tests. Unless otherwise specified, the reference rounds shall be assembled in accordance with the requirements of Drawing 11825947 and inspected in accordance with DOD-C-63980 with the charge weight controlled to ± 0.2 grams.

4.5.12 Ballistic reference rounds. A minimum of 10 reference rounds shall be fired during each ballistic firing test and the mean muzzle velocity, case mouth pressure and action time recorded. Any individual round readings which vary from the mean by greater than 12 meters per second, 20 MPa or 0.3 milliseconds respectively, shall be disregarded and a new mean value shall be computed. If more than three rounds are disregarded, additional reference rounds shall be fired, provided they are fired during the same day as the applicable tests are conducted. The differences between the mean reference round test values and the reference acceptance baseline values shall not exceed 10 meters per second, 23 MPa, or 0.35 millisecond. Reference round data shall be utilized as defined in 4.5.22.

4.5.13 Accuracy reference rounds. The accuracy reference baseline shall be established during lot acceptance testing of the reference lots in accordance with the test of Table III, except a minimum of 60 rounds, divided into 20 round groups, shall be used to establish the baseline. The accuracy of each 20 round group shall be determined and an average calculated for the baseline. Reference rounds shall be utilized as defined in 4.5.22.

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4.5.14 Penetration test.

4.5.14.1 Dynamic penetration test. The target shall be 25.4MM nominal thickness rolled homogeneous armor plate. The plate shall be positioned 500 meters ± 10 meters from the muzzle at an angle of obliquity equal to 50° ± 2° from the nominal. The test barrel conforming to drawing 9390748 shall be used in this test.

4.5.14.2 Static penetration test. Test procedure shall be submitted by the contractor to the PCO for approval. Test procedure shall call for detonating 25 projectiles at each spin rate against a 3" thick RHA target at a stand-off distance which simulated dynamic impact. Failure to meet the requirements of 3.14.2 shall result in referring the lot. Disposition will then be based on performing the test of 4.5.14.1 to insure compliance with 3.14.1.

4.5.15 Normal impact function. The target shall be placed 200 meters ± 20 meters from the muzzle. The failure of 11 or more projectiles to function high order detonation shall be cause for rejection of the lot. If 3 - 10 failures occur, the lot will be retested with the double sampling plan:

		n_1	=90	c_1	=2
r_1	=11				
		n_t	=270	c_t	=10
r_{1t}	=11				

If two or less failures occur during the initial test sample, the lot will be accepted. The test barrel conforming to Drawing 9390748 shall be used in this test.

4.5.15 Normal impact function. The target shall be placed 200 meters ± 20 meter from the muzzle. Failure of any cartridges to function high order detonation shall be cause for rejection of the lot.

4.5.16 Disassembly torque test of fuze on projectile. After four (4) hours curing time, the projectile shall be placed in a fixture and the Newton/Meter (inch pounds) requirement specified on the applicable drawing shall be applied. Observation shall be made of subject requirement. This test is a non-destructive test.

4.5.17 Propellant contamination. After completion of test specified in 4.5.8, each of the cartridges tested will be examined visually for contamination as described in AMCR 715-505, Chapter 1, Section 10, Volume 8.

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4.5.18 Retest procedures. Failure of the materiel sample to comply with the velocity/pressure and action time requirements shall result in a retest for the failed requirements. Sample size shall be as indicated in the applicable table.

4.5.19 Accuracy retest criteria. If the Mann barrel dispersion standard deviation (test) exceeds the requirements specified in 3.11 the following procedure shall be followed:

a. Fire a 20 round accuracy test utilizing reference TP ammunition for which the Mann barrel dispersion baseline is known. Calculate the radial standard deviation of the TP rounds (TP_{ref}).

b. Calculate the difference between the TP reference baseline and a 20 round test group.

$$TP_{ref} - TP_{baseline} = TP$$

c. Correct to standard deviation of the test ammunition as follows:

NOTE: corrected = test - TP

4.5.20 Temperature tolerences. Unless otherwise specified all temperatures are tolerenced $\pm$ 5°C.

4.5.21 Calibration of instrumentation. A reference lot of 30MM TP ammunition will be manufactured. It will be tested and established as a control baseline. The reference lot accuracy will be established by averaging the first three-20 round accuracy tests. Prior to acceptance testing of each lot, a minimum of ten rounds per day will be fired to establish pressure, velocity and action time. Ballistic variation between reference and acceptance tests lots will be adjusted as follows:

$$x_a + (X_{ra} - X_r) = \text{corrected } X_a$$

Where X_a is mean of acceptance lot

X_r is mean of reference lot.

X_{ra} is mean of acceptance baseline.

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5. PACKAGING

5.1 Packaging and packing. The cartridge shall be preserved, packaged and packed in accordance with the applicable requirements for the shipment of hazardous items as defined in Title 49, Code of Federal Regulations, parts 100 through 199 and in accordance with the applicable drawings.

5.1.1 Packing.

5.1.1.1 Linked packing. The linked pack shall consist of linked cartridges packed in the M592 shipping and storage container, (dwg. 10542565) and may include provisions for interconnecting ammunition belts. The links shall conform to Drawing 11825879. The packing ratio of left to right belts shall be specified in the contract or purchase order.

5.1.1.2 Bulk packing. The bulk pack shall consist of cartridges packed in the M592 shipping and storage container.

5.1.1.3 Bulk (Metallic Strip) packing. The Bulk (Metallic Strip) Pack shall consist of 10 30mm Metallic Ammunition Strips (see drawing 12902719) containing 11 cartridges each packed in the M592 shipping and Storage container.

5.1.2 Ammunition container interior packing.

5.1.2.1 Linked packing. The interior packing shall be in accordance with Drawing 11825891 or 11825892 as applicable for right hand or left hand pack.

5.1.2.2 Bulk packing. The interior packing shall be in accordance with Drawing 11825890.

5.1.2.3 Bulk (Metallic Strip) packing. The interior packing shall be in accordance with drawing number 12902718. The 30mm rounds shall be placed in the Metallic Strips in accordance with drawings 1291089.

5.2 Marking. Marking for shipment shall be in accordance with the instructions of the contract and shall comply with drawing number 11825893 (for Bulk "Metallic Strip" Packing, marking shall comply with drawing number 12902718). In addition, exterior containers shall show the caution markings required by Title 49, Code of Federal Regulations, indicating the appropriate classification of ammunition.

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5.3 Documentation with shipment. Each shipment of ammunition shall contain data cards in accordance with MIL-STD-1167, lot number assignment in accordance with MIL-STD-1168, data of manufacture and complete definition of the ammunition including such items as case, primer, flashtube assembly, propellant, charge weights, projectile and lot identification.

5.4 Shipping. When shipments of more than one lot are made at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers assembling and packing for Cartridge, 30MM, M789 for firing in the M230, the Aden MK IV and DEFA series of guns.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

6.3 Process deviation. A process deviation is defined as a change in the approved method of manufacture or an operational change which may alter the metallurgical or physical properties of the item.

6.4 Visual defects standards. Components and assemblies which have been rejected during initial production shall be segregated and visual standards established jointly by representatives of the Government and the contractor. Subsequent application of these standards for acceptance inspection shall be specified in the contract or purchase order.

6.5 Accuracy measurement procedure.

a. Select a reference point on the target and establish X and Y Axis through this point. Measure X and Y distance to each impact point.

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b. Calculate mean impact point (X, Y), (i.e., $\bar{X} = \sum X_i/N, \bar{Y} = \sum Y_i/N$).

c. Calculate standard deviations for X and Y.

$$x(m) = \frac{(X_i - \bar{X})^2}{N-1}, \quad y(m) = \frac{(Y_i - \bar{Y})^2}{N-1}$$

6.6 Mann barrel. The barrel used for preproduction and lot acceptance testing shall not be used for testing if any of the following conditions exist.

a. The barrel does not meet drawing requirements.

b. Degradation of reference lot muzzle velocity by more than 15 meters per second.

If accuracy degradation in excess of 0.4 mil exists, the barrel shall be replaced.

6.7 Standard deviation. The standard deviation for use in testing for projectile extraction, muzzle velocity, case mount pressure and action time shall be calculated in accordance with the following formula:

$$= \frac{(x_i - \bar{x})^2}{n-1}$$

6.8 Minimum cartridge weight. The minimum cartridge weight shall be determined for each lot based on the minimum propellant charge weight selected per 4.5.9. The formula for minimum propellant cartridge weight, when assembled in accordance with Drawing 11825954 is defined below:

Minimum weight (grams) - 292 grams + charge weight (grams)

6.9 Muzzle velocity correction sample. A velocity of 800.2 meters per second is recorded in a test instrumentation configuration with the mid-point between the velocity coils located 30 meters from the muzzle. The muzzle velocity is calculated as follows:

Muzzle Velocity - 800.2 + (0.75 x 30) = 823 meters per second

6.10 Data reporting. Final data calculations shall be rounded off as follows:

Velocity	-	Nearest meter per second
Pressure	-	Nearest 0.1 MPa

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Action time - Nearest 0.01 ms

Accuracy - Nearest 0.01 milliradian

6.11 Gun stoppage criteria. A gun stoppage is defined as failure to complete a single round cycle.

6.12 Projectile coating defect criteria. Projectiles indicating exposure of bare metal or underlying phosphate or organic pre-treatment shall be considered unsatisfactory if any of the following conditions exist.

a. Any area exceeding 60 square millimeters.

b. More than one area exceeding 30 square millimeters or less.

c. More than five areas of 15 square millimeters or less.

d. Missing paint on the rotating band is not a defect.

6.13 Rotating band failure criteria. Rotating band failure shall be defined as loss of any portion of the band which exceeds a 13MM length and has evidence of band seat serration imprint, or the wiping down to the bourellet diameter (no engraving in evidence) over the total length of the rotating band. For classification as a defect, there must be evidence, either by microflash photographs, recovery of the band or portions thereof, or by hole(s) in the witness board/screen. Normal band fringing is acceptable.

6.14 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I (D), Picatinny Arsenal, N.J. 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract.

6.15 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankfort Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.16 Subject Term (Key word Listing).

Cannon Caliber Ammo
Helicopter
Projectile
Cartridge Case

6.17 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.18 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 1305-AE20)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revis letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification c requirements on current contracts. Comments submitted on this form do not constitute or imply authorizatior waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-63982A (AR)	920826

3. DOCUMENT TITLE: CARTRIDGE, 30MM HIGH EXPLOSIVE DUAL PURPOSE, M789

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: 201-724-6675 (2) AUTOVON: DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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